Synthetic Cultural Agents as Scoring Instruments: A Preregistered Held-Out Evaluation of Survey Item Screening

Abstract

Synthetic respondents built from large language models promise a cheap first reading of survey items, yet are rarely tested as screening instruments on items they were not built for. We evaluate synthetic cultural agents, which compile the signs of declared country-level preference benchmarks into parameter adapters through a regularized pairwise choice model, as a scoring instrument for new item text. Under a preregistered protocol, the adapters alone assigned a dimension and direction to candidate World Values Survey items before any human value was read. In a retrospective screen against the existing survey, selected items predict the direction of cross-national differences better than random assignment, but the signal is weak, and no dimension passes both a second-bank check and the screen. A factorial over sign profiles shows entangled dimensions, more so under prompting. Because the human criterion is retrospective, pre-piloting remains a prospective use that this paper does not test.

Keywords: synthetic respondents; large language models; survey item screening; preregistration; cross-national measurement; direct preference optimization.

1 Introduction

Cross-national survey programs choose their items long before they know how those items will behave. A candidate question about trust in institutions or willingness to take risks must be translated, pretested, and fielded in many countries before a researcher can see whether its country-level differences line up with the construct it was meant to capture. Language models offer a tempting shortcut: a researcher supplies a question and a description of a population and reads off synthetic responses (Argyle et al., 2023; Horton et al., 2023; Buskirk et al., 2026). The same literature documents why that shortcut is fragile. Synthetic responses are sensitive to prompt wording, compress the variance of human answers, and can misportray and flatten identity groups (Bisbee et al., 2024; Boelaert et al., 2025; Wang et al., 2025), and the alignment procedures that make models usable change how faithfully they simulate population attitudes (Lyman et al., 2025). Methodological work in this journal concludes that validation against human data remains essential, and proposes designs that treat model predictions as potentially informative observations beside a human gold standard (Kozlowski and Evans, 2025; Broska et al., 2025).

Most of that empirical work asks whether a model reproduces the answers people gave to questions whose answers were already known. Survey design poses a different question. Before fielding, the useful property of an instrument is whether it can tell a researcher which of many new items are likely to separate countries in the intended direction, when the instrument was fixed before those items existed. That is a screening problem, and a screening instrument has to be judged on items it was not built for, with a decision rule fixed in advance and a human criterion read only afterwards. It also has to be judged against cheaper alternatives, because a screen

that does no better than a careful reading of the questionnaire adds cost without adding information.

We study one such instrument. Gonzalez-Bonorino et al. (2026) proposed synthetic cultural agents: group-indexed choice policies built by compiling the signs of declared country-level preference benchmarks from the Global Preferences Survey (GPS; Falk et al., 2018) into parameter adapters. Each country's sign profile deterministically labels a shared bank of paired synthetic responses, and Direct Preference Optimization (Rafailov et al., 2023), read as a regularized Bradley–Terry pairwise choice model, fits one adapter per profile. The adapters are evaluated with prompts that never name a country. On a sixteen-country development panel, that construction carried the coarse GPS trust sign onto new World Values Survey (WVS) item wording, while agreement with human responses remained unresolved and survey pre-piloting was left as a prospective use. The construction is inspectable: the researcher controls which aggregate moments enter, the sign rule that turns them into labels, and whether the group is named at evaluation.

This paper asks what that construction is worth as a scoring instrument for new item text, under conditions designed to make the answer hard to flatter. We trained adapters for every country in the intersection of the GPS and WVS Wave 7 rather than a selected panel, and a second set of adapters for all 64 possible sign profiles of the six GPS dimensions on a newly generated scenario bank. We then ran three evaluations. The first asks whether the trained surfaces of the original battery survive the replacement of the scenario bank. The second is a preregistered screen of 223 WVS Wave 7 items that no adapter was built for: a probe that reads only the adapters' scores assigns each item a preference dimension and a direction, a fixed rule selects the items with the strongest assignments, and only then are the selected items compared with the human country differences. The third uses the full factorial of sign profiles as

a diagnostic of whether the six declared dimensions behave as separate inputs, and compares the trained adapters with a prompt that states the same six signs.

The results are mixed, and we report them as they came out. Trust on the original battery carries over to the second scenario bank; for risk taking, positive reciprocity and altruism, the agreement with the GPS keeps a positive point estimate but its interval now includes zero. In the retrospective screen of held-out items, the probe-selected items predict the direction of human country differences better than random assignment of the same dimensions and signs, and the screen therefore rejects its preregistered null. The signal is weak: it is a small fraction of the agreement that two waves of the same human survey achieve on the same items, and its bootstrap interval reaches zero. Within the screen, only the items assigned to risk taking stand apart from zero, and that dimension does not survive an exploratory adjustment for the four dimensions tested; the items assigned to positive reciprocity point the wrong way. On the few items where they overlap, the registered contrasts do not distinguish the adapters from a careful reader of the questionnaire. In the factorial, every sign bit moves its own surface in the intended direction, but the bits also move each other's surfaces, often by more, and the planned check of separability fails. Prompting with the same signs is more entangled still and moves three surfaces the wrong way. No preference dimension passes both the second-bank check and the held-out screen.

Two features of the design bound what these results can say. The human criterion is retrospective: the WVS Wave 7 file existed before the probe, the item pool and the adapters, and it is the only human source. The screen therefore tests whether the instrument's selections agree with an existing survey; it cannot show that screening would change which items a researcher fields. And the instrument is built from signs alone, so it carries no information about magnitudes or about variation among people within a country.

Our contributions are three.

1. **The instrument at full scale.** We extend the sign compiler from a selected panel to every GPS–WVS country and to the complete factorial of sign profiles on a second scenario bank, so that the construction can be read as a designed manipulation of synthetic training labels.

2. **A preregistered retrospective screening protocol.** We specify, register, and run a screen of new survey items with a probe fixed before any human value was read, a permutation null that absorbs the probe's own selection rate, a human ceiling from two survey waves, and two comparators that read only the questionnaire.

3. **A diagnostic of entanglement.** We use the sign-profile factorial to show which declared dimensions the instrument keeps apart and which it confounds, and we compare trained adapters with prompting on the same information.

Together these contributions describe what the sign compiler holds, where it fails, and what remains untested when it is used as a scoring instrument for item text. The intended use is group-level: a reusable, inspectable instrument that a survey designer can consult beside human pretesting, and whose readings concern differences among countries rather than the answers of any person. Section 2 places the instrument among current uses of synthetic respondents and in the measurement literature on cross-national comparability. Section 3 describes the construction, Section 4 the evaluation design and its registration, Section 5 the results, and Section 6 what they permit a researcher to do.

2 Synthetic Respondents as Measurement Instruments

Synthetic respondents and their critics. The proposal that a language model conditioned on a description of a population can stand in for a sample of its members has two well-known forms. In silicon sampling, the description enters the prompt and the model's answer distribution is read as an estimate of the population's (Argyle et al., 2023); in the homo silicus program, the model plays an economic agent in a stylized decision problem (Horton et al., 2023). A systematic review maps the rapid spread of these uses across survey research (Buskirk et al., 2026). Where prompted models have been compared with human surveys, their answers are sensitive to prompt wording and unstable over time (Bisbee et al., 2024), and they show a strong bias and low variance on each topic, with a bias that changes unpredictably from one topic to the next (Boelaert et al., 2025). Prompting with group labels can also flatten and misportray the groups it names (Wang et al., 2025). Fine-tuning on survey responses is one answer to these problems (Suh et al., 2025; Cao et al., 2025; Krsteski et al., 2026), and the choice between aligned and unaligned models changes how faithfully a model simulates population attitudes (Lyman et al., 2025). Two reviews in this journal draw the methodological conclusion that matters here. Kozlowski and Evans (2025) identify features of current models that impair the realistic simulation of human subjects and conclude that validation against human data remains essential. Broska et al. (2025) propose a mixed subjects design in which model predictions are potentially informative observations and human subjects remain the gold standard. Both positions require a criterion against which the synthetic reading can fail.

Construct, measure, and anchor. Synthetic respondents are easier to evaluate when three objects are kept apart. A construct is a theoretical preference attribute, such as generalized trust. A measure operationalizes it through observed answers or choices, such as the response distribution to a WVS question about whether most people can be trusted. An anchor is an aggregate score that conditions the instrument, such as a country's GPS trust coordinate. The GPS supplies such anchors for six economic preferences (trust, risk taking, patience, altruism, positive reciprocity and negative reciprocity), measured with experimentally validated survey modules in a common design across countries (Falk et al., 2018). We treat these coordinates as declared aggregate moments: the researcher chooses them, and nothing about the instrument's construction certifies that they are the right moments for a given survey. Mapping a WVS item to a GPS dimension makes the item a candidate indicator of that dimension, and an average of such items is a proposed composite until its interpretation is supported. Agreement between an instrument and an anchor is therefore a different fact from agreement between an instrument and people, and agreement between a survey and an anchor across countries is a third. The multitrait–multimethod logic of Campbell and Fiske (1959) applies directly: the human survey is the other method, and an instrument that agrees with its own anchor has passed a construction check without yet meeting a criterion.

Cross-national comparability. The unit of analysis in this paper is the country. Country-level agreement does not establish that an item measures the same construct for individuals in different countries, and the literature on measurement equivalence explains why: loadings and intercepts can differ across groups in ways that leave country means comparable for some purposes and misleading for others (Steenkamp and Baumgartner, 1998; Davidov et al., 2014). Recent work in this journal argues

for a pragmatic approach in which composite scores aggregated to the country level can suffice for simple group comparisons despite noninvariance, while warning that such scores can understate uncertainty (Bratt, 2025), and shows by simulation that unaddressed measurement error and, to a lesser degree, noninvariance bias multilevel estimates (Pokropek et al., 2025). We adopt the same posture. The screen in this paper asks a group-level directional question about country rankings, and it makes no claim about invariance or individual-level validity.

Pairwise choice and preference optimization. The instrument rests on a correspondence between a training objective for language models and a classical model of paired comparisons. Under the Bradley and Terry (1952) model, the probability that one option is chosen over another is a logistic function of the difference in their latent utilities, which is the binary logit of random-utility theory. Direct Preference Optimization fits a policy to labeled pairs by maximizing exactly this likelihood, with the utility index written as a scaled log ratio between the fitted policy and a fixed reference policy (Rafailov et al., 2023). The scale parameter penalizes divergence from the reference, so the fitted policy is a regularized Bradley–Terry estimator. The synthetic labels in our application are deterministic; we use the likelihood as a fitting criterion and make no claim that it models human choice noise.

The empty cell. Prior evaluations of synthetic respondents score models on questions whose human answers are known, usually with the population named in the prompt, and fine-tuning approaches learn from human survey responses. What is missing is a test of a fixed synthetic-respondent instrument as a screen: an instrument built from declared aggregate inputs alone, applied to items it was not built for, with its selection rule and inference registered before the human criterion is read, and

compared with screens that read only the questionnaire. Section 4 describes such a test. Whether the instrument passes it is a question for Section 5.

3 The Instrument

The instrument maps declared aggregate preference signs into choice policies and reads those policies on survey items without naming any country. We summarize the construction of Gonzalez-Bonorino et al. (2026) and state what this paper changes. Throughout, a language model is treated as a conditional distribution over response sequences given a prompt (Ludwig et al., 2026).

3.1 From declared signs to a shared pair bank

Let $z_c = (z_{c1}, \ldots, z_{cK})$ be the GPS coordinates of country c on the K dimensions trust, risk taking, patience, altruism, positive reciprocity and negative reciprocity. Only the sign profile s_c, with $s_{ck} = 1\{z_{ck} \geq 0\}$, enters the construction. A teacher model and a response generator produce a shared bank of scenarios, each with two responses intended to express a higher and a lower level of one dimension k_i. For country c, the chosen and rejected responses on pair i are

$$(y_{w,ci},\, y_{\ell,ci}) = \begin{cases} (y_i^+,\, y_i^-), & s_{c,k_i} = 1, \\ (y_i^-,\, y_i^+), & s_{c,k_i} = 0. \end{cases} \tag{1}$$

The scenario and response text is identical for every country; only the label changes. No scenario names a country and none reuses WVS question wording, and the intended loading of each pair is a generator judgment rather than a verified trait manipulation.

Two countries with the same sign profile receive identical labels, so the instrument cannot distinguish them, and nothing in the construction carries the magnitude $|z_{ck}|$.

3.2 Preference optimization as regularized pairwise choice

Each sign profile is fitted with Direct Preference Optimization. Under the Bradley–Terry model, the probability that y_w is chosen over y_ℓ after prompt x is $\sigma\{r(x, y_w) - r(x, y_\ell)\}$, with σ the logistic function and r a latent utility index. DPO writes the index as $r(x, y) = \beta \log\{\pi_\theta(y \mid x)/\pi_{\text{ref}}(y \mid x)\}$ up to a prompt-specific constant that cancels from pairwise differences, and minimizes the negative log likelihood of the labeled pairs,

$$\mathcal{L}_c(\theta) = -\frac{1}{|\mathcal{D}_c|} \sum_{(x, y_w, y_\ell) \in \mathcal{D}_c} \log \sigma \left\{ \beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_\ell \mid x)}{\pi_{\text{ref}}(y_\ell \mid x)} \right\}. \tag{2}$$

The parameter β penalizes departures from the reference policy, which makes the estimator a regularized Bradley–Terry fit (Bradley and Terry, 1952; Rafailov et al., 2023). Because the labels are deterministic, the likelihood serves as a fitting criterion and says nothing about the noise in human choices. Each adapter is a low-rank update to a quantized Llama 3.1 8B Instruct base model (Hu et al., 2022; Dettmers et al., 2023), trained jointly on all six dimensions for one epoch on 526 of the 658 pairs, with LoRA rank 16 and $\beta = 0.1$. A joint adapter does not isolate any single dimension; whether the dimensions separate in practice is the subject of the factorial diagnostic in Section 4.

3.3 Two cohorts and a prompt route

The paper uses two sets of adapters. The first cohort has one adapter for each of the 42 countries in the intersection of the GPS and WVS Wave 7, trained on the original scenario bank. These 42 countries realize 26 distinct sign profiles. The second cohort is trained on a second, independently generated bank of 658 pairs under the same sign rule and the same text restrictions, and it contains one adapter for every one of the 64 possible sign profiles of the six dimensions. Because the labels depend on the country only through its sign profile, each of the 42 countries takes the adapter of its profile when country-level comparisons are needed; we verified that each country's labels on the second bank equal those of its profile. The 64 profiles form a full 2^6 factorial in the six sign bits. Each profile is a constructed design cell, and most of them are realized by no country.

The sign-profile prompt route gives the base model the same six signs in words, with no country name and no adapter. It carries the same declared information through a different channel, which makes it the natural comparison for the trained route. It is different from the country-named persona prompt of Gonzalez-Bonorino et al. (2026), which also changes the information supplied.

3.4 Readout

For an item q with candidate response codes j, the instrument computes the teacher-forced log probability of each code after the rendered item prompt and normalizes over the supplied codes,

$$\widehat{P}_u(j \mid q) = \frac{\exp \ell_{uqj}}{\sum_h \exp \ell_{uqh}}, \qquad y_{uq} = \sum_j g_q(j)\, \widehat{P}_u(j \mid q), \tag{3}$$

where ℓ_{uqj} is the summed token log probability of code j for unit u (an adapter or a prompted profile) and g_q is the item's recode. The score y_{uq} is the expected recoded response. No evaluation prompt names a country. The prompt template, the option order, the leading space before each code and the recode are frozen in a scoring specification whose hash every scoring run must reproduce before a model is loaded. On the original 23-item WVS battery, item scores are averaged into seven surfaces: trust, risk taking, altruism, positive reciprocity, negative reciprocity, and two separate patience items. We call this the strict battery. The normalized code probabilities are a scoring convention, and their dispersion need not match the response distributions of any human population.

4 Evaluation Design

The evaluation separates what each comparison permits. A construction check asks whether the adapters encode what they were trained on. A second-bank check asks whether that encoding survives a change of scenario text. A held-out screen asks whether the instrument's readings of new items agree with an existing human survey. A factorial diagnostic asks whether the declared dimensions behave as separate inputs. We report every number with a label for the surface it was computed on (Table 1), and numbers from different surfaces are never pooled.

4.1 Second-bank check

For each surface of the strict battery, we compute the Spearman correlation between the adapters' surface composites and the GPS coordinate of the surface's own dimension over the 42 countries, first for the first cohort (S1) and then for the second cohort with each country assigned its profile's adapter (S2). Intervals come from a profile-block

Table 1: Evaluation surfaces. Every number in the paper carries one of these labels.

Label	Units and items	Comparison	What it can show
S1	First-cohort country adapters (original bank), strict battery	Adapter–GPS rank correlation over countries; model side only	Whether a surface encodes its declared sign
S2	Second-cohort profile adapters (second bank), each country mapped to its profile, strict battery	Same statistic and bootstrap as S1	Whether that encoding survives a new scenario bank
H	Second-cohort adapters on held-out WVS Wave 7 items	Signed agreement of selected items with human country means, against GPS	Retrospective screen of new items against an existing survey
F	Second-cohort adapters and prompted profiles, strict battery	Main effects of the six sign bits on seven surfaces; model side only	Whether the declared dimensions separate
R	One profile retrained without a fixed seed	Score differences against the original adapter; model side only	Retraining noise for one unit
N	Two near-zero refits and size-matched controls	Composite shifts in profile standard deviations; model side only	Sensitivity of the sign rule's margin

bootstrap that resamples the 26 sign profiles realized by the countries, with 2,000 draws, seed 20261011, and percentile limits. The first-cohort values reproduce the banked run exactly. The imprint set is the 4 dimensions whose first-cohort interval excluded zero: trust, risk taking, positive reciprocity and altruism. Patience and negative reciprocity were flat in the first cohort and are reported without interpretation. The rule was fixed before the second cohort was trained: an imprint dimension whose second-bank interval includes zero is named, and no sentence about that dimension carrying over to the second bank is written. We also report the agreement between the two banks over the 26 profiles they share, as the Pearson correlation between second-bank profile composites and first-bank means over the countries in each profile, with a 2,000-draw bootstrap over profiles. Bank-to-bank agreement measures how much the scenario text changes the instrument; it involves no human data.

4.2 The held-out screen

Item pool. The pool was built from WVS Wave 7 metadata alone (variable labels, value labels and questionnaire text). An item enters if it is a substantive question outside the battery used to develop the adapters, below the demographic block, with two to eleven contiguous ordinal response codes. A text review excluded items whose code order is not monotone or that test factual knowledge, confirmed the ordinal reading of others, and repaired wording that did not extract from the questionnaire, recording a reason for every change. The result is 223 items that no adapter was built for.

Probe. All 64 second-cohort adapters scored every pool item under the frozen specification. For each item i, the probe regresses the 64 adapter scores on an intercept, the six sign bits coded ± 0.5, and their 15 two-way interactions, which

leaves 42 error degrees of freedom. With t_d the main-effect t statistic of dimension d, the item's assigned dimension is $d_i = \arg\max_d |t_d|$, or none when the largest $|t_d|$ is below 2.0, and its sign is $s_i = \text{sign}(b_{d_i})$. The primary rule keeps items whose assigned dimension is in the imprint set and whose $|t|$ is at least 2, up to the top 25 percent of all pool items by $|t|$, which caps the selection at 56 items. The secondary rule keeps every imprint item above the fixed threshold; it was declared a reliability check and does not enter the inference. The probe reads model scores only. It was stated in advance that, under pure noise, about a quarter of items cross the threshold on at least one dimension; the selection rule does not control this rate, and the permutation null below absorbs it.

Statistic and criterion. For each selected item, the signed item agreement is

$$a_i = s_i \cdot \rho\big(\bar{h}_{ci},\, z_{c,d_i}\big), \qquad A = \frac{1}{|S|} \sum_{i \in S} a_i, \tag{4}$$

where ρ is the Spearman correlation over the 42 countries, $\bar{h}_{ci}$ is the survey-weighted WVS Wave 7 mean of item i in country c with negative codes set to missing, and S is the selected set. An item–country cell that was never fielded enters as missing and is not imputed. A is positive when the human country differences on the selected items run in the direction the instrument assigned, against the GPS coordinate of the assigned dimension.

Inference. The registered test is a permutation null over items: draw as many items as were selected uniformly from the pool, give them the selected items' dimension and sign pairs in random order, and recompute A; 10,000 draws, seed 20261015, one-sided p equal to the share of null draws at or above the observed A. Because the null redraws items from the same pool, it absorbs the rate at which the probe flags noise items. A

95 percent interval for A comes from 2,000 bootstrap resamples of the 42 countries with percentile limits. The registered readings were fixed in advance: if A is not above the null at one-sided $\alpha = 0.05$, the adapters do not screen held-out items better than random assignment; if A is above the null but below one third of the human ceiling, a screening signal exists but is weak.

Human ceiling. For each selected item fielded in both WVS waves 6 and 7, we computed the Spearman correlation between the two waves' country means over the 28 countries with both waves (World Values Survey Association, 2022) and averaged it over items. This is the agreement two human surveys reach on these items. The ratio of A to the ceiling is a benchmark only, because the two statistics are built differently: A correlates one item with an anchor over 42 countries, while the ceiling correlates one item with itself across waves over fewer countries.

Per-dimension readings. The registered analysis also reports A_d, the mean of a_i over the selected items assigned to dimension d, with an interval from the same 2,000 country resamples, and the share of items with $a_i > 0$. An imprint dimension whose interval includes zero is read as giving no evidence of agreement for that dimension. These intervals are unadjusted for testing four dimensions. As an exploratory analysis added after the registered results were read, we report Bonferroni intervals over the 4 imprint dimensions, read off the same bootstrap draws at the 98.75 percent level.

Comparators. Two comparators read only the questionnaire. The face-valid rule assigns an item to a dimension when exactly one dimension's construct words occur in its wording and reads the direction from the response labels; a person could only confirm, flip or abstain, with a written reason. The second comparator is a typed decision model that returns calibrated probabilities (jev-1.13.0), given the item wording,

the response options and the six GPS construct definitions used to generate the training data; because single calls were unstable near the decision threshold, the comparator of record is 5 independent runs, and an item is assigned only when at least 3 runs agree on dimension and sign. Neither comparator saw any model score or human response. The registered primary contrast is coverage-matched: $D_{\text{common}} = A_{\text{adapter}} - A_{\text{comparator}}$ on the items both the adapter selection and the comparator assign, with an interval from the paired country bootstrap. An interval above zero would mean the adapters find structure the wording does not reveal; an interval that covers zero was registered as "the adapters match a careful reader"; an interval below zero would favor the reader's rule. The contrast on each route's own items is reported beside it. Both comparators concentrate on trust, which the preregistration named in advance as a limitation.

Readout variants and the prompt route. The probe was re-estimated on three re-scorings of the pool: options in reversed order, and two paraphrases of the prompt template. The registered statistic is the share of selected items whose dimension and sign are unchanged; the selection was declared dependent on wording if more than half change. The same probe and test were also applied to the 64 prompted sign profiles.

4.3 Registration, blinding and amendments

The held-out screen was registered in a document committed to the project repository and timestamped by the pushed git tag `prereg-heldout-2026-10-14`, before any held-out item was scored by an adapter. At that point no selection existed and no held-out human value had been read; the only WVS Wave 7 content read was metadata and questionnaire text. A freeze guard enforced the order of work. After the probe selection, a freeze file recorded the hashes of the selection and the item pool, and

every program that reads held-out human data refuses to run unless that file exists and both hashes still match. The order was: score the held-out items, run the probe, freeze, and only then join the human data. The freeze was committed a few minutes before the first read of held-out human data.

Two amendments were made, both before any human held-out value was read (Figure 1). The first scoring pass ran on the pool as registered. Reading the item text afterwards found 12 items whose standalone wording could not be answered as written (sentences cut off mid-clause, a stripped country placeholder, grid rows truncated to a pronoun, and follow-ups that refer to a question absent from a standalone prompt). Amendment 1 repaired these items from their own response options or variable labels, under the pool rule and override mechanism already registered, and every adapter, readout variant and prompted profile was re-scored under the amended specification as a single run. We disclose that the first probe pass had been computed and read before the repairs were chosen, and that two of the repaired items (Q153 and Q157) were in its top quartile. The comparators had assigned all pool items on the pre-repair wording. Amendment 2 removed the repaired items from both comparator routes before any comparator statistic was computed; this affected one item (Q82), which was not in the adapter selection. The decision was taken with the instruction to freeze, before any human value was read; the amendment text was written after the primary statistic was read and records that earlier decision.

The criterion is retrospective. The WVS Wave 7 file is public and existed before the probe, the pool and the adapters, and it is the only human source in this study. Registration fixed the analysis before the human values were read by us, which guards against tuning the probe to the criterion. It cannot make the criterion prospective, and the screen tests agreement with an existing survey. Results on surface H are therefore labelled retrospective throughout. Every scoring run reproduced its pinned

specification hash before a model was loaded. All manifests of the held-out runs record one code revision; launcher files around it were patched in place during a recovery, and the scoring modules are byte-identical between the patched setup and the recorded revision. The strict-battery run of the second cohort spans two recorded revisions. The online supplement lists the revisions, their differences in the readout path and the file hashes, and it reports two provenance checks. The commit that produced the first cohort's scores cannot be named, and a replay traced the dominant difference between those scores and an unquantized replay to the four-bit quantization the adapters were trained on. A report-only comparison of base-model scores with the archived scores of the earlier study shows a gap whose remaining cause is not established, and no statistic in this article uses those archived scores.

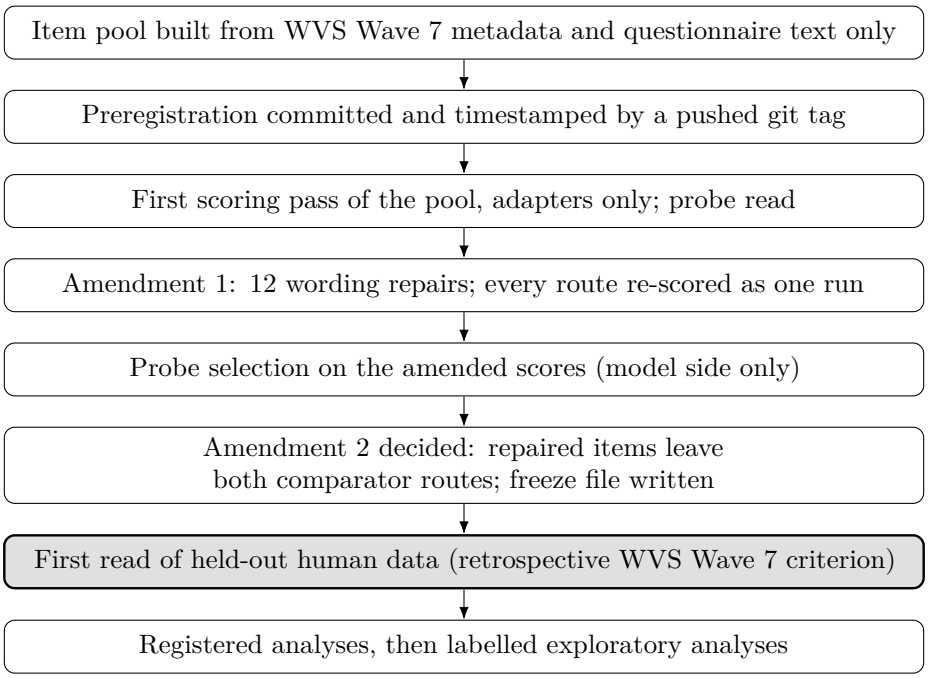


Figure 1: Order of work for the held-out screen. Every step above the shaded box used model scores, metadata or questionnaire text only. The shaded box is the single read of held-out human data.

4.4 The sign-profile factorial

Because the second cohort covers all 64 sign profiles, the strict-battery composites of its adapters form a full factorial in six binary factors. For each of the seven surfaces we fit the same regression as the probe (intercept, six sign bits coded ± 0.5, 15 two-way interactions, 42 error degrees of freedom). A main effect is then the difference between the mean composite of profiles with the bit set and profiles without it. We call it the main effect of the sign bit on the adapter's scores. This is a causal statement about the instrument, because the bits were assigned by design. The manipulation acts on synthetic training labels, and its effects are changes in the model's response distributions. They carry no implication about how preferences operate in people.

Each bit has an own surface (patience owns two), so there are seven own effects. The project's analysis plan, separate from the held-out registration, set one acceptance check for the factorial: the factorial is informative about separability if at least 4 of the six bits move their own surface more than any other surface, a criterion we call V5, with a binomial reference that takes the chance of a hit as one in seven. Because raw effects compare surfaces with different response scales, we add two scale-free views as exploratory analyses: the same comparison on effects divided by each surface's standard deviation over the 64 profiles, and a column-wise count of surfaces whose largest mover among the six bits is their own bit. We also report the share of variance of the standardized composites on their first principal component and the mean absolute correlation between surfaces. The same analyses on the 64 prompted profiles give the route contrast, which compares two instruments carrying the same declared information.

4.5 Retraining noise and the near-zero margin

One profile adapter was retrained under the same protocol as an unseeded retrain, providing a noise-floor estimate for the pipeline. The low-rank initialization of the training code is not seeded, so the retrain shares the recipe and the data and differs in its random initialization. We compare its scores with the original on both batteries.

The sign rule treats a coordinate just above zero like a large positive one. A sensitivity check fixed before any refit was trained removes the training pairs of every dimension whose GPS coordinate lies within 0.10 of zero and refits two predeclared countries (Great Britain and Colombia), each beside a size-matched control trained on the same number of pairs drawn at random. The contrast of interest is the refit minus its control on the dimensions still in training, in units of the surface's standard deviation over the 64 profiles, read against the retrain's shift on the same surface as a noise floor. The predeclared rule was to report the margin beside the main results and never to promote it.

5 Results

We report the second-bank check first, because it says what the instrument encodes before any human data enter. The retrospective screen follows, then its comparators and readout checks, then the factorial diagnostic and the noise checks. Each number carries the label of its surface.

5.1 A second scenario bank

The first cohort encodes four of the six declared signs on the strict battery (S1). Over the 42 countries, the adapter–GPS Spearman correlation is 0.744 [0.500, 0.865] for

trust, 0.435 [0.085, 0.686] for risk taking, 0.488 [0.120, 0.695] for positive reciprocity, and 0.406 [0.020, 0.676] for altruism. These four are the imprint set that the held-out screen later uses. They are surfaces of a new cohort on a new pipeline stage, and they are not comparable term for term with the sixteen-country development panel of Gonzalez-Bonorino et al. (2026).

When the scenario bank is replaced (S2, first column of Table 2), trust keeps its agreement with the GPS, at 0.803 [0.615, 0.876]. The other three imprint dimensions keep positive point estimates, but their intervals now include zero, so 1 of the 4 imprint dimensions passes the second-bank check and 3 do not. Under the rule fixed before the second cohort was trained, we write no sentence about risk taking, positive reciprocity or altruism carrying over to the second bank. The surfaces that were flat on the first bank stay flat on the second: negative reciprocity 0.239 [−0.237, 0.604], patience Q43 0.005 [−0.403, 0.470], and patience Q50 0.161 [−0.343, 0.557].

The two banks agree closely on most surfaces when compared profile by profile. Over the 26 shared profiles, the Pearson correlation between first-bank and second-bank composites is 0.909 [0.797, 0.965] for trust, 0.949 [0.859, 0.989] for altruism, 0.898 [0.775, 0.960] for positive reciprocity, 0.708 [0.488, 0.861] for negative reciprocity, and 0.598 [0.229, 0.819] for risk taking. Risk taking is the least stable surface across banks, which fits its loss on the second-bank check. For altruism and positive reciprocity the profile composites of the two banks are highly correlated, and the point estimates of agreement with the GPS fall only modestly; their intervals cross zero because they were already wide. Bank-to-bank agreement measures how much the scenario text changes the instrument. It says nothing about people.

5.2 The retrospective screen

Retrospectively, the screen rejects its registered null. The probe selected 56 of the 223 pool items under the primary rule. Their mean signed agreement with the WVS Wave 7 country differences is $A = 0.061$, with a country-bootstrap interval of $[-0.007, 0.126]$. Under the permutation null, which reassigns the same dimension and sign pairs to items drawn at random from the pool, the null distribution has mean 0.012 and ninety-fifth percentile 0.046, and the one-sided p is 0.010. Of the selected items, 60.7% have a positive item agreement. The human data contain 174 item–country cells that were never fielded; they enter as missing.

By the registered reading, the signal is weak. Two WVS waves agree with each other on the selected items fielded in both waves with a mean Spearman correlation of 0.746 (median 0.767, 34 items). The screen's A is 8.1% of that ceiling, well below the registered bar of one third, which would be 0.249. The ratio is a benchmark, because the two statistics are built differently. The bootstrap interval reaches zero while the permutation test rejects the null, and the two answer different questions. The permutation test asks whether the instrument's assignments beat random assignments of the same pairs to items from the same pool, and it absorbs the rate at which the probe flags noise items. The bootstrap asks how much A would vary across samples of countries. The direction of the finding is therefore clearer than its size.

The registered reliability checks point the same way. The secondary rule, which keeps every imprint item above the fixed threshold, gives $A = 0.070$ over 185 items, with a one-sided permutation $p < 0.001$ (null ninety-fifth percentile 0.034). Dropping one country at a time moves the primary A between 0.051 and 0.072. In an exploratory check on the 34 items fielded in both waves, A is 0.043 against wave-6 country means

and 0.047 against wave-7 means over the 28 countries with both waves, so the small signal does not depend on one wave.

Table 2: The second-bank check and the retrospective screen, by imprint dimension. The two checks are reported side by side and never averaged.

Dimension	S2: second bank ρ [95% interval]	H: primary rule A_d [95% interval], items	H: primary rule, Bonferroni A_d [adjusted interval]	H: secondary rule A_d [95% interval], items
Trust	0.803 [0.615, 0.876]	0.062 [−0.047, 0.173], 12	[−0.088, 0.199]	0.092 [−0.019, 0.199], 44
Risk taking	0.210 [−0.199, 0.584]	0.158 [0.015, 0.290], 13	[−0.026, 0.326]	0.143 [0.045, 0.229], 50
Positive reciprocity	0.366 [−0.068, 0.652]	−0.071 [−0.281, 0.154], 11	[−0.332, 0.205]	−0.049 [−0.198, 0.117], 22
Altruism	0.364 [−0.057, 0.653]	0.069 [−0.036, 0.177], 20	[−0.058, 0.198]	0.040 [−0.038, 0.117], 69

Note: S2 is the Spearman correlation between second-bank adapter composites and the GPS coordinate over 42 countries, with a profile-block bootstrap (2,000 draws, seed 20261011); model side only. H is the retrospective screen against WVS Wave 7 country means: A_d is the mean signed item agreement over the selected items the instrument assigned to the dimension, with the registered country bootstrap (2,000 resamples, seed 20261015). The registered intervals are unadjusted. The Bonferroni intervals over the 4 imprint dimensions are exploratory and read off the same draws. The secondary rule is a registered reliability check. No dimension has an interval above zero on both S2 and H.

The per-dimension readings show where the signal sits (Table 2). Under the primary rule, only the items assigned to risk taking have an interval above zero, $A_d = 0.158$ [0.015, 0.290] on 13 items, of which 92.3% agree in sign. The intervals for trust and altruism include zero. The items assigned to positive reciprocity point the wrong way: their point estimate is negative, and only 18.2% of the 11 items agree in sign. The secondary rule agrees on both points, with risk taking alone above

zero at 0.143 [0.045, 0.229] over 50 items and a negative point estimate for positive reciprocity.[1]

The registered per-dimension intervals are unadjusted for testing four dimensions. In the exploratory Bonferroni adjustment, read off the same bootstrap draws, the primary-rule interval for risk taking becomes [−0.026, 0.326] and includes zero. The one primary-rule dimension that excludes zero therefore does not survive adjustment for four dimensions. Under the secondary rule the adjusted risk-taking interval stays above zero, [0.017, 0.253].

Read together, the two checks in Table 2 disagree by dimension. Trust is the dimension that survives the second bank, and its retrospective screen interval includes zero. Risk taking is the dimension whose retrospective screen interval excludes zero before adjustment, and it is one of the three that lose their interval on the second bank. No imprint dimension passes both checks. The per-dimension cells are small, and a margin like the one for risk taking on 13 items is one dimension out of four on one pool.

5.3 Questionnaire readers and the prompt route

The registered contrasts do not distinguish the adapters from a careful reader of the questionnaire (Table 3). Every interval covers zero, which the preregistration reads as "the adapters match a careful reader". That reading rests on few items. The coverage-matched contrasts use the 5 items that the adapters and the face-valid rule both assign and the 9 items they share with the typed decision model, and the routes agree on dimension and sign for 1 and 7 of them. An interval that covers zero on so few items is consistent with a match and does not establish one. Both readers

[1]The probe's per-dimension permutation p values were not registered. For altruism under the primary rule they point the other way from the registered interval, so we report the registered intervals only.

Table 3: Retrospective screen: the adapters against two questionnaire readers and against prompting with the same signs.

	Face-valid reader	Typed decision model, majority of 5 runs	Sign-profile prompt
Items assigned by both routes	5	9	16
Same dimension and sign on those items	1	7	6
Adapter minus route, shared items	0.020 [−0.181, 0.215]	−0.002 [−0.064, 0.061]	0.004 [−0.065, 0.075]
Adapter minus route, own items	−0.034 [−0.242, 0.169]	−0.067 [−0.237, 0.103]	0.033 [−0.083, 0.143]
Route's own A (items)	0.095 (25)	0.128 (40)	0.028 (56)

Note: Differences are adapter minus the named route, with 95% intervals from the paired country bootstrap (2,000 resamples). The registered primary contrast with each questionnaire reader is the one on shared items. The questionnaire readers' own items are restricted to the imprint set, with the 12 items repaired by Amendment 1 removed. The adapter's own A is 0.061 on 56 items. Every interval in the table covers zero.

concentrate on trust, which the preregistration named in advance. Their own-item A is higher than the adapters' overall A and higher than the 0.062 of the adapters' own trust items, while the per-dimension cells of each reader hold too few items to be informative.

Prompting the base model with the same six signs does not pass the registered null. The same probe applied to the 64 prompted profiles gives $A = 0.028$ [−0.109, 0.158] with a one-sided permutation p of 0.509, and it assigns its selected items to a single dimension, trust. The adapter route passes the null and the prompt route does not, but the direct difference between them is unresolved: adapter minus prompt is 0.033 on each route's own items and 0.004 on the 16 items both select, and both intervals cover zero. The two routes agree on dimension and sign for 6 of those shared items.

5.4 Readout robustness

Item-level assignment is stable under option order and paraphrase, whereas membership in the particular selected set churns. The registered statistic, the number of selected items whose dimension and sign are unchanged, is 39 of 56 with reversed options, 39 with the first paraphrase, and 30 with the second. No variant changes more than half, so the registered condition for wording dependence is not met, although the second paraphrase comes close. Membership is a different matter, because the primary rule is a top-quartile cut on $|t|$ and small shifts reorder items near the cut. Only 18, 29 and 16 of the 56 items are selected under both the original readout and the respective variant, and the correlation of $|t|$ between the original and each variant over the pool is 0.176, 0.546 and 0.271. A researcher who reads the instrument should therefore trust the dimension and direction it assigns to an item more than the particular list of items it ranks highest.

5.5 The sign-profile factorial

On the trained route, every sign bit moves its own surface in the intended direction (Table 4). All seven own effects are positive, from 0.364 for trust ($t = 16.74$) to 0.124 for positive reciprocity ($t = 3.02$), and 0 of 7 point the wrong way.

The planned separability check fails. Only 2 of the 6 bits, those for altruism and negative reciprocity, move their own surface more than any other surface, where at least 4 were required (binomial $p = 0.207$). The scale-free views agree: 2 of 6 on standardized effects, and 1 of 7 surfaces, trust, whose largest mover is its own bit. Each bit moves its own surface and also moves others, often by more. The risk-taking bit moves the altruism surface by 0.983, against the altruism bit's own effect of 0.242, and the negative-reciprocity surface by -0.668, against an own effect of 0.625. The

Table 4: Own effects of the six sign bits and summary of entanglement, by route. Strict battery, all 64 sign profiles.

	Trained adapters		Sign-profile prompt	
Own effect on surface	Main effect	t	Main effect	t
Trust	0.364	16.74	1.780	394.46
Negative reciprocity	0.625	11.64	−0.023	−1.80
Altruism	0.242	6.32	0.364	11.99
Patience (Q50)	0.133	6.12	0.421	17.63
Risk taking	0.140	4.14	−0.457	−22.31
Patience (Q43)	0.049	3.63	−0.069	−19.75
Positive reciprocity	0.124	3.02	0.103	5.66
Summary of entanglement	Trained adapters		Sign-profile prompt	
Own effects against the intended direction	0 of 7		3 of 7	
Bits dominant on their own surface (V5)	2 of 6		2 of 6	
Same, on standardized effects	2 of 6		1 of 6	
Surfaces whose largest mover is their own bit	1 of 7		1 of 7	
Surfaces whose largest mover is the trust bit	3 of 7		7 of 7	
Variance on the first principal component	51.3%		83.5%	
Mean absolute correlation between surfaces	0.395		0.807	

Note: Main effects are differences in mean surface composite between profiles with the bit set and without it, from an OLS fit with the six bits and their 15 two-way interactions (42 error degrees of freedom); composites are expected recoded responses averaged over the surface's items, so the units differ across surfaces. The manipulation is of synthetic training labels or prompted signs; the effects are changes in model response distributions. V5 was a planned acceptance check (at least 4 of 6 bits); the other summary rows are exploratory.

trust bit moves patience Q50 by 0.495 and positive reciprocity by 0.418, against own effects of 0.133 and 0.124, and it is the largest mover on 3 of the 7 surfaces. One common direction carries 51.3% of the variance of the seven standardized surfaces, and the mean absolute correlation between surfaces is 0.395. The sign bits are therefore inputs that the instrument keeps distinct in direction and mixes in magnitude.

The prompted route fails the separability check by the same count, 2 of 6 (binomial $p = 0.207$), so V5 does not separate the routes. The common direction and the sign of the own effects do. Under prompting, the first principal component carries 83.5% of the variance, the mean absolute correlation between surfaces is 0.807, and the trust bit is the largest mover on all 7 surfaces. The prompted trust bit alone moves risk taking by -0.858 and patience Q50 by 1.774. Three prompted own effects point against the intended direction: risk taking at -0.457 ($t = -22.31$), patience Q43 at -0.069 ($t = -19.75$), and negative reciprocity at -0.023, which at $t = -1.80$ is not distinguishable from zero. The contrast compares two instruments that carry the same declared information, one trained and one prompted. It concerns entanglement and the direction of own effects, and it is silent on screening accuracy, where the difference between routes is unresolved. The 64 profiles are constructed design cells, and the country files realize far fewer of them.

5.6 Retraining noise and the near-zero margin

The unseeded retrain of one profile reproduces the original adapter's readings closely. On the strict battery (140 scored cells), the largest single-cell difference in code probability is 0.052 and the mean absolute difference is 0.0060; the Pearson correlation of expected codes over the 23 items is 0.9998. On the held-out pool (1,087 cells), the corresponding values are 0.092, 0.0056 and 0.9996 over 223 items. Every strict-surface

shift is at most 0.112 of a standard deviation across profiles, with patience Q50 the largest. This bounds retraining noise for one unit; the variance of retraining across units was not measured.[2]

6 Discussion

We set out to evaluate the sign compiler as a scoring instrument for new item text, and the evaluation returns a specific profile of what holds, what fails, and what remains untested. We take the three contributions in turn and then state the limits that apply to all of them.

The instrument. Trained at full scale, the instrument encodes the direction of its declared signs: on the trained route, every sign bit moves its own surface the intended way, although only some of those surfaces agree with the GPS across countries (Section 5.1). The magnitude of the encoding is fragile for most dimensions. Only trust keeps its agreement with the GPS when the scenario bank is replaced, and the bank-to-bank comparison locates risk taking as the dimension that the scenario text moves most (Section 5.1). A researcher can therefore treat the instrument's trust surface as a property of the construction that survives a change of generator, and should treat the other surfaces as properties of a particular bank until a further bank

[2]Near-zero margin (surface N). Under the predeclared rule the near-zero filter is not promoted. On a retained imprint surface, Colombia's positive reciprocity moved 0.092 profile standard deviations against its intended direction relative to its control, beyond the declared replicate floor of 0.011. That move is much smaller than the variation between two small-data trainings: a size-matched control differs from the full-data baseline by up to 1.222 standard deviations for Great Britain and 1.499 for Colombia. We claim no reversal. Great Britain's trust moved 0.448 in its intended direction, which is also within that variation. On the dropped dimensions, 4 of 7 surfaces moved against their intended direction beyond the floor, as expected when a dimension has no training pairs; this is a loss-of-signal check and bears on neither side. Two refits are a screen, with no interval and no claim across countries. The declared floor comes from one unseeded retrain on the full data and is too small to read small-data refits against, as the controls show.

says otherwise. This is a statement about the instrument and its anchor. It is silent on whether the WVS items measure the GPS constructs in people.

The retrospective screen. The screen asked whether a probe that reads only the instrument picks items whose human country differences run in the assigned direction, and the registered answer is yes, weakly. The signal is a small fraction of what two human waves achieve on the same items, its bootstrap interval reaches zero, and it is concentrated in one dimension. That dimension, risk taking, is the one the second bank does not support, and its interval does not survive an exploratory adjustment for the four dimensions tested. Trust, the dimension that does survive the second bank, does not stand apart from zero in the screen. The items assigned to positive reciprocity point the wrong way. That disagreement could arise from the instrument, from the mapping between WVS items and the GPS construct, or from limited coherence between the two surveys for this dimension, and the design cannot separate these explanations. Read as cross-survey evidence, it marks positive reciprocity as the dimension on which the instrument's readings and the human survey part ways. On the items that the adapters share with two readers of the questionnaire, the registered contrasts cannot tell the adapters apart from those readers. The readers work mostly on trust and the overlap is small, so this comparison leaves open whether the adapters add anything a careful reader would miss on the dimensions the readers do not cover.

The diagnostic. The factorial shows that the declared dimensions are entangled inside the instrument. Each bit moves its own surface the intended way, but the bits also move one another's surfaces, and a single common direction carries about half of the variance across surfaces. The planned separability check fails on both routes. Prompting with the same signs is more entangled still: one common direction

dominates, the trust sign moves every surface, and three own effects point the wrong way. The same pattern appears on the held-out items, where the prompted probe assigned its selected items to trust alone and did not pass the null. Training on sign-labeled pairs is thus the route that keeps the direction of each declared input. It does not deliver separate dimensions. Because the two routes carry the same declared information, the contrast describes two instruments, and it makes no claim about which training recipe is best or about screening accuracy, where the difference between routes is unresolved.

Limits. Five limits apply to every result above. First, the human criterion is retrospective. The WVS Wave 7 file existed before the probe, the pool and the adapters, and it is the only human source. Registration and the freeze guard protect against tuning the probe to the criterion, but they cannot make the criterion independent of the survey program from which the item pool was drawn. Pre-piloting, in the sense that screening with the instrument would change which items a researcher fields relative to a human-only rule declared in advance, is a prospective use that this paper does not test. Second, the instrument is built from signs. It carries no information about magnitudes, and it cannot represent differences among countries that share a sign profile or among people within a country. Its readings are group-level, and country-level agreement does not establish individual-level validity or measurement invariance (Davidov et al., 2014; Bratt, 2025). Third, every adapter shares one base model, one training recipe and one readout convention, and the readout's code probabilities are a scoring convention whose dispersion need not match human response distributions. Fourth, the per-dimension cells are small and the selected set churns under readout variants, so the instrument's item-level assignments are more stable than any particular list of selected items. Fifth, the noise checks are narrow: one

unseeded retrain bounds retraining noise for one unit, and two near-zero refits are a screen with no claim across countries.

Intended use. Within those limits, the instrument has a modest and inspectable use. A survey designer can read the dimension and direction it assigns to candidate items as one input beside expert review and human pretesting, keeping in view that no dimension passed both checks. Its value lies in what it makes explicit: the declared moments, the sign rule, and the absence of any group name at evaluation are all on the record, so a reader can see what information the instrument was given and check where its readings depart from people. It complements fielding and pretesting, and the evidence here gives no reason to use it in their place. In the terms of Broska et al. (2025), its readings are potentially informative observations that still require a human criterion.

7 Conclusion

Synthetic cultural agents compile declared aggregate preference signs into choice policies that can be read on any survey item without naming a group. Used as a scoring instrument for new item text, they keep the direction of every declared sign on the strict battery, keep the trust encoding across a change of scenario bank, and produce a weak retrospective screening signal on held-out WVS items that exceeds random assignment. They do not keep the declared dimensions separate, no dimension passes both the second-bank check and the screen, and on the few items where comparison is possible the registered contrasts do not distinguish them from a careful reader of the questionnaire. Prompting a model with the same signs is more entangled than training and moves three of its own surfaces the wrong way.

A researcher may use the instrument now as an inspectable, group-level reading of candidate items beside human pretesting. The test that would license more is prospective: probe-selected items fielded to people after the selection and a human-only comparison rule have both been declared, so that the instrument's selections can be compared with what a researcher would otherwise have fielded. Until then, the instrument complements human pretesting and fielding.

Preregistration Statement

The held-out screen (Sections 4.2 and 5.2) was preregistered. The registration document was committed to the project repository and timestamped by the git tag `prereg-heldout-2026-10-14`, pushed before any held-out item was scored by an adapter. Amendment 1 (repair of 12 items whose standalone wording could not be answered) was made after the first model-side probe pass had been computed and read, and before any human value was read. The freeze file was committed, and Amendment 2 (removal of those items from both comparator routes) was decided, before the single read of held-out human data; the text of Amendment 2 was written afterwards and records that earlier decision. The criterion data, WVS Wave 7, are public and existed before the study, so the registration fixes the analysis before the authors read the human values and the criterion remains retrospective. All registered analyses of the screen are reported in Section 5.2 and Tables 2 and 3, except two descriptives (item-level agreement after dropping items with small $|a_i|$, and the face-valid reader's statistic on all of its items including negative reciprocity), which are reported in the online supplement. The Bonferroni intervals, the wave-specific screen statistics, the scale-free factorial views and the factorial route contrast are exploratory. The factorial's separability check (V5) was fixed in the project's analysis

plan, separately from the registration. The second-bank decision rule was fixed before the second cohort was trained and was not part of the registration. An anonymized copy of the registration is provided with the submission.

Code, Materials, and Data Availability

Analysis code, the item pool and its overrides, the frozen selection, the comparator files, all model scores, and the scripts that build every number in this article from those scores are available for review at an anonymized link. Model adapters and score archives are released at acceptance. The frozen inputs are identified by SHA-256 hashes: the amended item pool `19241f6f103c51d6`... as written to disk, whose line-ending-normalized form in the repository is `a1dbceea1a33da05`...; the overrides file `0f5cd2bbd88ef6ea`...; and the frozen selection `782b7d2123f9758e`.... The pool is regenerated byte for byte from the overrides file by the pool-building script; full hashes and the line-ending convention are in the online supplement. WVS and GPS data are available from their providers (Haerpfer et al., 2022; World Values Survey Association, 2022; Falk et al., 2018) and are not redistributed; no respondent-level data or country-level human means are included in the replication materials.

Institutional Review

This study analyzes publicly available, de-identified secondary survey data and involves no interaction with human participants.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Declaration of Generative AI Use

Large language models are the object of study in this article. In addition, an AI coding and writing assistant was used under the authors' direction to write analysis code and to draft and edit manuscript text; the authors reviewed all code, numbers and text and take responsibility for them.

References

Argyle, L. P., E. C. Busby, N. Fulda, J. R. Gubler, C. Rytting, and D. Wingate (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis 31*(3), 337–351.

Bisbee, J., J. D. Clinton, C. Dorff, B. Kenkel, and J. M. Larson (2024). Synthetic replacements for human survey data? the perils of large language models. *Political Analysis 32*(4), 401–416.

Boelaert, J., S. Coavoux, É. Ollion, I. Petev, and P. Präg (2025). Machine bias. how do generative language models answer opinion polls? *Sociological Methods & Research 54*(3), 1156–1196.

Bradley, R. A. and M. E. Terry (1952). Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika 39*(3/4), 324–345.

Bratt, C. (2025). Benefits of a pragmatic approach: Rethinking measurement invariance and composite scores in cross-cultural research. *Sociological Methods & Research*. Online first.

Broska, D., M. Howes, and A. van Loon (2025). The mixed subjects design: Treating large language models as potentially informative observations. *Sociological Methods & Research 54*(3), 1074–1109.

Buskirk, T. D., F. Keusch, L. von der Heyde, and A. Eck (2026). More parameters than populations: A systematic literature review of large language models within survey research.

Campbell, D. T. and D. W. Fiske (1959). Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological Bulletin 56*(2), 81–105.

Cao, Y., H. Liu, A. Arora, I. Augenstein, P. Röttger, and D. Hershcovich (2025). Specializing large language models to simulate survey response distributions for global populations. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Albuquerque, New Mexico, pp. 3141–3154. Association for Computational Linguistics.

Davidov, E., B. Meuleman, J. Cieciuch, P. Schmidt, and J. Billiet (2014). Measurement equivalence in cross-national research. *Annual Review of Sociology 40*, 55–75.

Dettmers, T., A. Pagnoni, A. Holtzman, and L. Zettlemoyer (2023). QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems*, Volume 36.

Falk, A., A. Becker, T. Dohmen, B. Enke, D. Huffman, and U. Sunde (2018). Global evidence on economic preferences. *The Quarterly Journal of Economics 133*(4), 1645–1692.

Gonzalez-Bonorino, A., K. Biriukova, and M. Capra (2026). Synthetic cultural agents from aggregate anchors. arXiv preprint 2610.06562.

Haerpfer, C., R. Inglehart, A. Moreno, C. Welzel, K. Kizilova, J. Diez-Medrano, M. Lagos, P. Norris, E. Ponarin, B. Puranen, et al. (2022). World values survey wave 7 (2017–2022) cross-national data-set. Madrid, Spain and Vienna, Austria: JD Systems Institute and WVSA Secretariat. Version 4.0.0.

Horton, J. J., A. Filippas, and B. S. Manning (2023, April). Large language models as simulated economic agents: What can we learn from homo silicus? Working Paper 31122, National Bureau of Economic Research. Revised February 2026.

Hu, E. J., Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen (2022). LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Kozlowski, A. C. and J. Evans (2025). Simulating subjects: The promise and peril of artificial intelligence stand-ins for social agents and interactions. *Sociological Methods & Research 54*(3), 1017–1073.

Krsteski, S., G. Russo, S. Chang, R. West, and K. Gligorić (2026). Valid survey simulations with limited human data: The roles of prompting, fine-tuning, and rectification. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, pp. 10887–10906. Association for Computational Linguistics.

Ludwig, J., S. Mullainathan, and A. Rambachan (2026). Large language models: An applied econometric framework. *Annual Review of Economics 18*, 283–316.

Lyman, A., B. Hepner, L. P. Argyle, E. C. Busby, J. R. Gubler, and D. Wingate (2025). Balancing large language model alignment and algorithmic fidelity in social science research. *Sociological Methods & Research 54*(3), 1110–1155.

Pokropek, A., T. Żółtak, E. Davidov, B. Meuleman, and P. Schmidt (2025). Challenges in multilevel modelling: Cross-group measurement noninvariance and measurement errors. a monte carlo simulation study. *Sociological Methods & Research*. Online first.

Rafailov, R., A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, and C. Finn (2023). Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, Volume 36.

Steenkamp, J.-B. E. M. and H. Baumgartner (1998). Assessing measurement invariance in cross-national consumer research. *Journal of Consumer Research 25*(1), 78–107.

Suh, J., E. Jahanparast, S. Moon, M. Kang, and S. Chang (2025). Language model fine-tuning on scaled survey data for predicting distributions of public opinions. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 21147–21170. Association for Computational Linguistics.

Wang, A., J. Morgenstern, and J. P. Dickerson (2025). Large language models that replace human participants can harmfully misportray and flatten identity groups. *Nature Machine Intelligence 7*, 400–411.

World Values Survey Association (2022). World values survey time-series (1981–2022) data file. Data file version 5.0.